Hints: Linear algebra – 8. Block matrices and determinants of special shape

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Column j of AB is $\sum_k b_{kj} A_k$, where A_k are the columns of A. Expand $\det(AB)$ by multilinearity in the columns: a sum over maps $j \mapsto k_j$.

Hint 2. Non-injective maps give repeated columns and contribute 0. Group the injective ones by their image S. The signs assemble into $\det A_{[m],S} \cdot \det B_{S,[m]}$.

Hint 3. For a tall $n \times m$ matrix A, apply this to $A^T \cdot A$: $\det(A^T A) = \sum_S (\det A_{S,[m]})^2$. Geometrically, $\det(A^T A)$ is the squared m-volume of the parallelepiped spanned by the columns. The identity is Pythagoras: it equals the sum of the squared volumes of its projections to the coordinate m-planes.

Problem 2

Hint 1. If $A_n x = \lambda x$, then (x, x) and $(x, -x)$ are eigenvectors of A_{n+1} with eigenvalues $\lambda + 1$ and $\lambda - 1$. This accounts for the whole spectrum of A_{n+1}.

Hint 2. By induction, the eigenvalues of A_n are the sums of n signs ± 1: the value $n - 2j$ occurs with multiplicity $\binom{n}{j}$, $j = 0, \dots, n$. (A_n is the adjacency matrix of the n-cube.)

Problem 3

Hint 1. The answer is $n_k = 2^k$. Example: $A_i = E \otimes \cdots \otimes N \otimes \cdots \otimes E$, with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ in the i-th factor. Their product is $N^{\otimes k} \neq O$.

Hint 2. Lower bound by induction on k: the A_i, $i < k$, commute with A_k, so they preserve $W = \operatorname{Im} A_k$. Their restrictions to W still square to zero and commute. Their product is nonzero on W, because $A_1 \cdots A_{k-1} A_k \neq O$.

Hint 3. So $\dim W \geqslant 2^{k-1}$. Also $A_k^2 = O$ gives $W \subseteq \ker A_k$, so $\dim W \leqslant n/2$.

Problem 4

Hint 1. View the determinant as a polynomial in x_n of degree $n - 1$. It vanishes at $x_1, \dots, x_{n-1}$, and its leading coefficient is the smaller Vandermonde determinant. This gives $\prod_{i<j}(x_j - x_i)$ by induction.

Hint 2. The Vandermonde matrix is the matrix of the evaluation map from polynomials of degree $< n$ to $\mathbb{C}^n$ at n points. Distinct points make it invertible, so the conditions are independent. Solving $Vc = y$ by Cramer's rule, or checking the formula directly, gives the Lagrange interpolation formula.

Problem 5

Hint 1. Multiply the determinant by $\prod_{i,j}(x_i + y_j)$. The result is a polynomial of degree $n^2 - n$ that is antisymmetric in the x's and in the y's. So it is a constant times $\prod_{i<j}(x_j - x_i)(y_j - y_i)$, which has the same degree.

Hint 2. Fix the constant, e.g. by induction: subtract the last row from the others and the last column from the others, and factor out. Alternatively compare the coefficients of a suitable monomial.

Hint 3. Hilbert matrix: $\frac{1}{i+j-1} = \int_0^1 t^{i-1}t^{j-1}\,dt$, the Gram matrix of independent monomials, hence positive definite. The formula with $x_i = i$, $y_j = j-1$ gives a ratio of products of factorials, roughly of size 4^{-n^2}.

Problem 6

Hint 1. Let $p(x) = \sum a_i x^i$. Since $B(AB)^j A = (BA)^{j+1}$, we get $BA\,p(BA) = B\,p(AB)\,A = O$.

Hint 2. Suppose $a_0 \neq 0$. Then 0 is not a root of p, so AB (annihilated by p) is invertible. Hence $m = \operatorname{rank} AB \leqslant n$, so $m = n$ and A, B are invertible square matrices.

Hint 3. Then BA is invertible too, and $BA\,p(BA) = O$ gives $p(BA) = O$. Contradiction.

Problem 7

Hint 1. (Putnam 2023 B6.) Count by $k = ai$: $s(i,j) = \#\{k \in [0,n] : i \mid k,\ j \mid n-k\}$. So $S = MN$, with M the $n \times (n+1)$ matrix $[\,i \mid k\,]$ $(k = 0,\dots,n)$ and N the $(n+1) \times n$ matrix $[\,j \mid n-k\,]$. Apply Cauchy–Binet: $\det S = \sum_k \det M^{(k)} \det N_{(k)}$, deleting column k of M and row k of N.

Hint 2. N is M^T with its rows reversed, so $\det N_{(k)} = \pm \det M^{(n-k)}$, with a sign depending only on n. For $k > n/2$, the row $i = k$ of $M^{(k)}$ has a single 1 (in column 0), and expanding gives $\det M^{(k)} = \pm 1$. For n even, $\det M^{(n/2)} = 0$ (two equal rows).

Hint 3. So each term is $\pm \det M^{(k)}$ with $k < n/2$. The resulting signed sum is evaluated by expanding the determinant of M with a row of ones added on top: it is 0, since it equals the row $i = 1$. Answer: $\det S = (-1)^{\lceil n/2 \rceil - 1} \cdot 2\lceil n/2 \rceil$. Check small n: $2, 2, -4, -4, 6, \dots$

Problem 8

Hint 1. Put $u_i = v_i - v_0$. The volume is $\sqrt{\det G}/n!$, where $G_{ij} = \langle u_i, u_j \rangle = \frac{1}{2}(d_{0i}^2 + d_{0j}^2 - d_{ij}^2)$.

Hint 2. Use row and column operations on the bordered matrix $\begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & (d_{ij}^2) \end{pmatrix}$: subtract the row and column of v_0 from the others, then clear the border. It reduces to $-2G$ bordered, which gives $\mathrm{vol}^2 = \frac{(-1)^{n+1}}{2^n (n!)^2} \det CM$.

Hint 3. Realisability: numbers d_{ij} on $n+2$ points come from points in $\mathbb{R}^n$ iff the matrix G built from them (with base point v_0) is positive semidefinite of rank $\leqslant n$. In particular the Cayley–Menger determinant of all $n+2$ points must vanish (zero $(n+1)$-volume), with the sign conditions from $G \succeq 0$ on the sub-simplices.

Problem 9

Hint 1. Expand along the first row: $D_n = 3D_{n-1} - D_{n-2}$, with $D_1 = 3$ and $D_2 = 8$ (and $D_0 = 1$).

Hint 2. Fibonacci numbers satisfy $F_{m+2} = 3F_m - F_{m-2}$. Compare initial values: $D_n = F_{2n+2} = \frac{\varphi^{2n+2} - \varphi^{-2n-2}}{\sqrt{5}}$, with $\varphi = \frac{1+\sqrt{5}}{2}$.

Problem 10

Hint 1. Let $g(x) = \sum_{k \geqslant 0} c_{k+1} x^k = 1 + 5x + 35x^2 + \dots$. The matrix is the Hankel matrix $(g_{i+j})_{i,j=0}^{n-1}$ of the coefficients of g. From the quadratic equation for $C = \sum c_k x^k$ derive $2xg^2 - (1-3x)g + 1 = 0$.

Hint 2. Hankel determinants are read off from the continued fraction (J-fraction) of the generating function: if $g = \cfrac{1}{1 - b_0 x - \cfrac{\lambda_1 x^2}{1 - b_1 x - \cfrac{\lambda_2 x^2}{\ddots}}}$, then $\det(g_{i+j})_0^{n-1} = \prod_{k=1}^{n-1} \lambda_k^{n-k}$.

Hint 3. Verify from the equation for g that $g = \frac{1}{1-5x-10x^2h}$ with $h = \frac{1}{1-7x-10x^2h}$. So all $\lambda_k = 10$, and $\det A = 10^{n(n-1)/2}$.